

we do for matrices with real entries. Recall that, in $\mathbb{R}^n$, the length of a vector $\mathbf{v}$ is $\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$. However, this dot product cannot be used as an inner product to define length of vectors in $\mathbb{C}$, because, if $\mathbf{v} = [1, i]$, we would have $\mathbf{v} \cdot \mathbf{v} = 0$. The fix for this problem is simple. Recalling that $|a + bi|^2 = (a - bi)(a + bi)$, we make an adjustment and define the *Euclidean inner product* in $\mathbb{C}^n$.

DEFINITION 9.1 Euclidean Inner Product

Let $\mathbf{u} = [u_1, u_2, \dots, u_n]$ and $\mathbf{v} = [v_1, v_2, \dots, v_n]$ be vectors in $\mathbb{C}^n$. The **Euclidean inner product** of $\mathbf{u}$ and $\mathbf{v}$ is

$$\langle \mathbf{u}, \mathbf{v} \rangle = \overline{u_1} v_1 + \overline{u_2} v_2 + \cdots + \overline{u_n} v_n.$$

Notice that Definition 9.1 gives $\langle [1, i], [1, i] \rangle = (1)(1) + (-i)(i) = 1 + 1 = 2$. Because $|a + bi|^2 = (a - bi)(a + bi)$, we see at once that, for $\mathbf{v} \in \mathbb{C}^n$, we have

$$\langle \mathbf{v}, \mathbf{v} \rangle = |v_1|^2 + |v_2|^2 + \cdots + |v_n|^2, \quad (1)$$

just as in the $\mathbb{R}^n$ case.

We list properties of the Euclidean inner product in $\mathbb{C}^n$ as a theorem, leaving the proofs as exercises. (See Exercises 16 through 19.)

THEOREM 9.2 Properties of the Euclidean Inner Product

Let $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ be vectors in $\mathbb{C}^n$, and let z be a complex scalar. Then

1. $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$, and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ if and only if $\mathbf{u} = \mathbf{0}$,
2. $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$,
3. $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$,
4. $\langle \mathbf{w}, \mathbf{u} + \mathbf{v} \rangle = \langle \mathbf{w}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{v} \rangle$,
5. $\langle z\mathbf{u}, \mathbf{v} \rangle = \overline{z} \langle \mathbf{u}, \mathbf{v} \rangle$, and $\langle \mathbf{u}, z\mathbf{v} \rangle = z \langle \mathbf{u}, \mathbf{v} \rangle$.

Property 1 of Theorem 9.2 and Eq. (1) preceding the theorem suggest that we define the **magnitude** or **norm** of a vector $\mathbf{v}$ in $\mathbb{C}^n$ as

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = \sqrt{\overline{v_1} v_1 + \overline{v_2} v_2 + \cdots + \overline{v_n} v_n}. \quad \text{Magnitude of } \mathbf{v}$$

Vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{C}^n$ are **perpendicular** (or **orthogonal**) if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$; property 2 of Theorem 9.2 tells us that $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ if and only if $\langle \mathbf{v}, \mathbf{u} \rangle = 0$. The vectors are **parallel** if $\mathbf{u} = z\mathbf{v}$ for some scalar z in $\mathbb{C}$. Notice that $\|z\mathbf{v}\| = |z| \|\mathbf{v}\|$, just as in the real case. Vectors of magnitude 1 are **unit vectors**. Having made these definitions, we will feel free to consider without further definition such things as orthogonal subspaces and orthonormal bases.

EXAMPLE 5 Find a unit vector in $\mathbb{C}^3$ parallel to $\mathbf{v} = [1, i, 1 + i]$.

SOLUTION Because $\|v\| = \sqrt{\langle v, v \rangle}$, we have

$$\|v\| = \sqrt{1(1) + (-i)i + (1-i)(1+i)} = \sqrt{1+1+2} = 2.$$

Either of the vectors $\pm \frac{1}{2}[1, i, 1+i]$ satisfies the requirement. ■

The Euclidean inner product given in Definition 9.1 reduces to the usual dot product of vectors if the vectors are in $\mathbb{R}^n$. However, we have to watch one feature very carefully:

The Euclidean inner product in $\mathbb{C}^n$ is not commutative.

(See Property 2 in Theorem 9.2.) For example,

$$\langle [1, i], [0, 1] \rangle = -i, \quad \text{but} \quad \langle [0, 1], [1, i] \rangle = i.$$

To illustrate the care that must be taken as a result of the noncommutativity of the inner product, we consider the Gram-Schmidt process applied in the vector space $\mathbb{C}^n$. Let u and v be vectors in $\mathbb{C}^n$. Then the vector

$$w = u - \frac{\langle v, u \rangle}{\langle v, v \rangle} v$$

is orthogonal to v , because

$$\langle w, v \rangle = \langle u, v \rangle - \frac{\langle v, u \rangle}{\langle v, v \rangle} \langle v, v \rangle = \langle u, v \rangle - \langle u, v \rangle = 0.$$

We must not use $u - \frac{\langle u, v \rangle}{\langle v, v \rangle} v$, whose inner product with v is generally not zero.

EXAMPLE 6 Transform the basis

$$\{[1, i, i], [1, 0, -i], [1, 0, 1]\}$$

of $\mathbb{C}^3$ to an orthonormal one, using the Gram-Schmidt process.

SOLUTION First we transform the given basis to an orthogonal one. Because $v_1 = [1, i, i]$ and $v_2 = [1, 0, -i]$ are orthogonal, we work with $v_3 = [1, 0, 1]$ and replace it by

$$\begin{aligned} v_3 - \frac{\langle v_1, v_3 \rangle}{\langle v_1, v_1 \rangle} v_1 - \frac{\langle v_2, v_3 \rangle}{\langle v_2, v_2 \rangle} v_2 \\ &= v_3 - \frac{1-i}{3} v_1 - \frac{1+i}{2} v_2 \\ &= [1, 0, 1] - \frac{1}{3}[1-i, 1+i, 1+i] - \frac{1}{2}[1+i, 0, 1-i] \\ &= \frac{1}{6}[1-i, -2-2i, 1+i]. \end{aligned}$$

In fact, we prefer to replace $\mathbf{v}_3$ by $[1 - i, -2 - 2i, 1 + i]$, which is just as good. An orthogonal basis is

$$\{[1, i, i], [1, 0, -i], [1 - i, -2 - 2i, 1 + i]\},$$

and an orthonormal basis is

$$\left\{ \frac{1}{3}[1, i, i], \frac{1}{2}[1, 0, -i], \frac{1}{2\sqrt{3}}[1 - i, -2 - 2i, 1 + i] \right\}. \quad \blacksquare$$

The Conjugate Transpose

In fixing up our old inner product to serve in $\mathbb{C}^n$ in Definition 9.1, we had to decide whether to take conjugates of the components u_i of the first vector in $\langle \mathbf{u}, \mathbf{v} \rangle$ or to take conjugates of the components v_i of the second vector. It is more convenient to take conjugates of components of the first vector for the following reason. The bulk of our work in $\mathbb{R}^n$ has been formulated in terms of column vectors, although we often use row notation to save space. For example, when working with matrix representations of linear transformations, we always write $A\mathbf{x}$, where $\mathbf{x}$ is a column vector. You may recall several instances where it has been convenient for us to use the fact that the inner product (dot product) of two vectors in $\mathbb{R}^n$ appears as sole entry in the matrix product of the first vector as a row vector and of the second vector as a column vector. That is, for column vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, we can write $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^T \mathbf{y}$, where $\mathbf{x}^T$ is a row vector and where no distinction is made between a 1×1 matrix and a scalar. Thinking in these terms, we can express the condition that the column vectors of a real square matrix A form an orthonormal basis as $A^T A = I$. In order to preserve these convenient algebraic formulas with as little change as possible, we choose to take conjugates of the components of the first vector in Definition 9.1, and we continue with a definition that allows us to recover these formulas.

DEFINITION 9.2 Conjugate Transpose, or Hermitian Adjoint

Let $A = [a_{ij}]$ be an $m \times n$ matrix with complex scalar entries.

1. The **conjugate** of A is the $m \times n$ matrix $\bar{A} = [\bar{a}_{ij}]$.
2. The **conjugate transpose** (or **Hermitian adjoint**) of A is the matrix $A^* = [\bar{a}_{ji}]^T$.

For column vectors $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$, we have

$$\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^* \mathbf{w}. \quad (2)$$

Moreover, the condition for an $n \times n$ complex matrix A to have orthogonal unit column vectors can be written as

$$A^* A = I.$$

EXAMPLE 7 Find the conjugate transpose A^* of the matrix

$$A = \begin{bmatrix} 1 & i & 1+i \\ 2 & 0 & i \\ 2i & 1 & 1-i \end{bmatrix}.$$

SOLUTION We form the transpose while taking the conjugate of each element, obtaining

$$A^* = \begin{bmatrix} 1 & 2 & -2i \\ -i & 0 & 1 \\ 1-i & -i & 1+i \end{bmatrix}.$$

Following are some properties of the conjugate transpose that can easily be verified. (See Exercise 32.)

THEOREM 9.3 Properties of the Conjugate Transpose

Let A and B be $m \times n$ matrices. Then

1. $(A^*)^* = A$,
2. $(A + B)^* = A^* + B^*$,
3. $(zA)^* = \bar{z}A^*$ for any scalar $z \in \mathbb{C}$,
4. If A and B are square matrices, $(AB)^* = B^*A^*$.

EXAMPLE 8 Using the properties in Theorem 9.3, show that, for any $n \times n$ matrix A , we have $(A + A^*)^* = A + A^*$.

SOLUTION Using properties 1 and 2 of Theorem 9.3, we have

$$(A + A^*)^* = A^* + (A^*)^* = A^* + A = A + A^*,$$

which is what we wished to show. ■

Unitary and Hermitian Matrices

Recall that a *real orthogonal matrix* is a square matrix having orthogonal unit column vectors. The complex analogue of such a matrix is known by another name.

DEFINITION 9.3 Unitary Matrix

A square matrix U with complex entries is **unitary** if its column vectors are orthogonal unit vectors—that is, if $U^*U = I$.

Our next example gives an important property of unitary matrices that is familiar to us from the real orthogonal case. Notice in this example how handy the notation v^*v is for $\langle v, v \rangle$.

EXAMPLE 9 Let $T: \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a linear transformation having a unitary matrix U as matrix representation with respect to the standard basis. Show that $\|T(\mathbf{z})\| = \|\mathbf{z}\|$ for all $\mathbf{z} \in \mathbb{C}^n$.

SOLUTION We know that $T(\mathbf{z}) = U\mathbf{z}$, because U is the standard matrix representation of T . Because $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle = \mathbf{v}^* \mathbf{v}$, we find by using property 4 of Theorem 9.3 that

$$\|U\mathbf{z}\|^2 = (U\mathbf{z})^* U\mathbf{z} = \mathbf{z}^* (U^* U) \mathbf{z} = \mathbf{z}^* \mathbf{z} = \|\mathbf{z}\|^2.$$

Taking square roots, we find that $\|U\mathbf{z}\| = \|\mathbf{z}\|$, so $\|T(\mathbf{z})\| = \|\mathbf{z}\|$. ■

Real symmetric matrices play an important role in linear algebra. We saw in Chapter 8 that they are very handy in work with quadratic forms. We also stated without proof in Theorem 5.5 that every real symmetric matrix is diagonalizable. Symmetric matrices are defined in terms of the transpose operation. The useful analogue of symmetry for matrices with complex scalars involves the conjugate transpose operation.

DEFINITION 9.4 Hermitian Matrix

A square matrix H is **Hermitian** if $H^* = H$ —that is, if H is equal to its conjugate transpose.

If a square matrix H actually has real entries, it is Hermitian if and only if it is symmetric. Notice that the condition $H = H^* = (H)^T$ implies that the entries on the diagonal of any Hermitian matrix are real numbers.

EXAMPLE 10 Example 8 shows that, for any square matrix A , the matrix $A + A^*$ is Hermitian. Illustrate that the entries on the diagonal of $A + A^*$ are real, using the matrix A in Example 7.

SOLUTION Example 7 shows that, for the matrix

$$A = \begin{bmatrix} 1 & i & 1+i \\ 2 & 0 & i \\ 2i & 1 & 1-i \end{bmatrix},$$

we have

$$A^* = \begin{bmatrix} 1 & 2 & -2i \\ -i & 0 & 1 \\ 1-i & -i & 1+i \end{bmatrix}.$$

Thus,

$$A + A^* = \begin{bmatrix} 2 & 2+i & 1-i \\ 2-i & 0 & 1+i \\ 1+i & 1-i & 2 \end{bmatrix},$$

which has real diagonal entries. ■

Hermitian matrices provide the proper generalization of real symmetric matrices to enable us to prove in the next section that every Hermitian matrix is diagonalizable. The special case of this theorem for real matrices thus finally provides us with a proof of the fundamental theorem that every real symmetric matrix is diagonalizable. The fundamental theorem for real symmetric matrices is tough to prove if we stay within the real number system, but it is a corollary of a fairly easy theorem for complex matrices. This illustrates how we can obtain true insight into theorems in real analysis and linear algebra by studying analogous concepts that use complex numbers.

SUMMARY

1. $\mathbb{C}^n$ is an n -dimensional complex vector space.
2. If $\mathbf{u} = [u_1, u_2, \dots, u_n]$ and $\mathbf{v} = [v_1, v_2, \dots, v_n]$ are vectors in $\mathbb{C}^n$, then $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{u_1}v_1 + \overline{u_2}v_2 + \dots + \overline{u_n}v_n$ is the Euclidean inner product of $\mathbf{u}$ and $\mathbf{v}$ and satisfies the properties in Theorem 9.2. In general, $\langle \mathbf{u}, \mathbf{v} \rangle \neq \langle \mathbf{v}, \mathbf{u} \rangle$.
3. For $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$, the vector $\mathbf{u} - \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v}$ is perpendicular to $\mathbf{v}$.
4. The conjugate transpose of an $m \times n$ matrix A is the $n \times m$ matrix $A^* = (\overline{A})^T$. The conjugate transpose operation satisfies the properties in Theorem 9.3.
5. A square matrix U is unitary if $U^*U = I$. A real unitary matrix is an orthogonal matrix.
6. A square matrix H is Hermitian if $H = H^*$. A real Hermitian matrix is a symmetric matrix.

EXERCISES

1. Explain how $\mathbb{C}^n$ can be viewed as a $2n$ -dimensional real vector space and as an n -dimensional complex vector space. Give a basis in each case.
2. Is it appropriate to view $\mathbb{R}^n$ as a subspace of $\mathbb{C}^n$? Explain.
3. Find AB and BA if

$$A = \begin{bmatrix} 1 & i & i \\ 1+i & 1 & -i \\ i & 1+i & 1-i \end{bmatrix} \text{ and}$$

$$B = \begin{bmatrix} -1 & 1+i & i \\ 2+i & i & 1-i \\ i & 1 & i \end{bmatrix}.$$

4. Find A^2 and A^4 if $A = \begin{bmatrix} 1 & 1+i \\ -1+i & i \end{bmatrix}$.

5. Find A^{-1} if $A = \begin{bmatrix} 1 & i \\ 1+i & 2+i \end{bmatrix}$.

6. Find A^{-1} if $A = \begin{bmatrix} i & 1+i \\ 1 & i \end{bmatrix}$.

7. Find A^{-1} if $A = \begin{bmatrix} 1 & i & 1-i \\ 1 & 1 & 1+i \\ 0 & -1+i & 1 \end{bmatrix}$.

8. Find A^{-1} if $A = \begin{bmatrix} i & 1-i & 1+i \\ 0 & 1 & i \\ 1-i & -i & 1-i \end{bmatrix}$.

9. Solve the linear system $A\mathbf{z} = \begin{bmatrix} i \\ 1+i \\ i \end{bmatrix}$ if A is

the matrix in Exercise 7.

10. Solve the linear system $Az = \begin{bmatrix} -1+i \\ 2+i \\ 1 \end{bmatrix}$ if A is

the matrix in Exercise 8.

11. Find the solution space of the homogeneous system

$$z_1 + iz_2 + (1-i)z_3 = 0$$

$$(1+2i)z_1 - z_2 + z_3 = 0$$

$$(1+i)z_1 + 2iz_2 + (3-2i)z_3 = 0.$$

12. Find the rank of the matrix

$$\begin{bmatrix} 1 & 1+i & i & 1-i \\ 1+i & 2+i & 1-i & 1 \\ 1+i & 1+4i & 1+3i & 2-i \end{bmatrix}.$$

13. Find the rank of the matrix

$$\begin{bmatrix} 2-i & i & 1-i & 1+3i \\ i & 1+i & -1+2i & i \\ 1-i & 1+2i & 1-3i & 2+3i \end{bmatrix}.$$

14. Find each of the indicated inner products.

a. $\langle 2+i, 2, i \rangle, [i, 1+i, 2-i]$ in $\mathbb{C}^3$

b. $\langle 1-i, 1+i, i \rangle, [1+i, 1-i]$ in $\mathbb{C}^2$

c. $\langle 1+i, 1, 1-i, 1 \rangle, [i, 1+i, i, 1-i]$ in $\mathbb{C}^4$

d. $\langle 2-i, 3+i, 1+i \rangle, [1+i, 2-i, 1+i]$ in $\mathbb{C}^3$

15. Compute $\langle u, v \rangle$ and $\langle v, u \rangle$ for each of the following.

a. $u = [1, i], v = [1+i, 1-i]$

b. $u = [1+i, 2-i, i], v = [1, 1-i, 1+i]$

16. Verify property 1 of Theorem 9.2.

17. Verify property 2 of Theorem 9.2.

18. Verify properties 3 and 4 of Theorem 9.2.

19. Verify property 5 of Theorem 9.2.

20. Find the magnitude of each of the following vectors.

a. $[1, i, i]$

b. $[1+i, 1-i, 1+i]$

c. $[1+i, 2+i, 3+i]$

d. $[i, 1+i, 1-i, i]$

e. $[1+i, 1-i, i, 1-i]$

21. Determine whether the given pairs of vectors are parallel, perpendicular, or neither.

a. $[1, i], [i, 1]$

b. $[1+i, i, 1-i], [1-i, 1, -1-i]$

c. $[1+i, 2-i], [3i, 3+i]$

d. $[1, i, 1-i], [1-i, 1+i, 2]$

e. $[1+i, 1-i, 1], [i, 1-i, -3-i]$

22. Find a unit vector parallel to

$$[1+i, 1-i, i].$$

23. Find a vector of length 2 parallel to

$$[i, 1-i, 1+i, 1-i].$$

24. Find a unit vector perpendicular to

$$[2-i, 1+i].$$

25. Find a vector perpendicular to both

$$[1, i, 1-i] \text{ and } [1+i, 1-i, 1].$$

In Exercises 26–29, transform the given basis of $\mathbb{C}^n$ into an orthogonal basis, using the Gram–Schmidt process.

26. $\{[1+i, 1-i], [1, 1]\}$ in $\mathbb{C}^2$

27. $\{[2+i, 1+i], [1+i, i]\}$ in $\mathbb{C}^2$

28. $\{[1-i, 1+i, 1+i], [1, 1, -1-i], [1, i, -i]\}$ in $\mathbb{C}^3$

29. $\{[1, i, i], [1+i, 1, i], [i, 1+i, 0]\}$ in $\mathbb{C}^3$

30. Find the conjugate transpose of each of the following matrices.

a. $\begin{bmatrix} 1 & 1+i & 2 \\ 1-i & 1+i & 2 \end{bmatrix}$

b. $\begin{bmatrix} i & 1+i \\ 2+i & 1-i \end{bmatrix}$

c. $\begin{bmatrix} 1+i & 1+i \\ 2-i & 1-i \\ 1 & 1-2i \end{bmatrix}$

d. $\begin{bmatrix} 1+2i & i & 1-i \\ i & 1-i & 1+i \\ 1+i & 2-i & 1+3i \end{bmatrix}$

31. Label each of the following matrices as Hermitian, unitary, both, or neither.

a. $\frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$

b. $\frac{1}{2} \begin{bmatrix} 1 & i & 1-i \\ -i & -2 & 1 \\ 1+i & 1 & 1 \end{bmatrix}$

c. $\frac{1}{\sqrt{6}} \begin{bmatrix} \sqrt{2} & \sqrt{2}i & \sqrt{2}i \\ \sqrt{3}i & 0 & \sqrt{3} \\ -i & -2 & 1 \end{bmatrix}$

d. $\frac{1}{4} \begin{bmatrix} 1 & i & 1-i \\ i & 3 & 1 \\ 1-i & 1 & 2 \end{bmatrix}$

32. Prove the properties of the conjugate transpose operation given in Theorem 9.3. [Hint: From Section 1.3, we know that

analogous properties of the *transpose* operation hold for real matrices and real scalars and can be derived using just field properties of $\mathbb{R}$, so they are also true for matrices with complex entries. Thus we can focus on the effect of the *conjugation*. From Theorem 9.1, we know that $\overline{z + w} = \bar{z} + \bar{w}$, $\overline{zw} = \bar{z}\bar{w}$, and $\bar{\bar{z}} = z$, for $z, w \in \mathbb{C}$. Use these properties of conjugation to complete the proof of Theorem 9.3.]

33. Mark each of the following True or False. Assume that all matrices and scalars are complex.

- a. The definition of a determinant, properties of determinants (the transpose property, the row-interchange property, and so on), and techniques for computing them are developed using only field properties of $\mathbb{R}$ in Chapter 4, and thus they remain equally valid for square complex matrices.
- b. Cramer's rule is valid for square linear systems with complex coefficients.
- c. If A is any square matrix and $\det(A) \neq 0$, then $\det(iA) \neq \det(A)$.
- d. If U is unitary, then $U^{-1} = U^T$.
- e. If U is unitary, then $(\bar{U})^{-1} = U^T$.
- f. The Euclidean inner product in $\mathbb{C}^n$ is not commutative.
- g. For $u, v \in \mathbb{C}^n$, we have $\langle u, v \rangle = \langle v, u \rangle$ if and only if $\langle u, v \rangle$ is a real number.
- h. For a square matrix A , we have $\det(\bar{A}) = \overline{\det(A)}$.
- i. For a square matrix A , we have $\det(A^*) = \det(A)$.
- j. If U is a unitary matrix, then $\det(U^*) = \pm 1$.

- 34. Prove that, for vectors $v_1, v_2, \dots, v_n$ in $\mathbb{C}^n$, $\{v_1, v_2, \dots, v_n\}$ is a basis for $\mathbb{C}^n$ if and only if $\{\bar{v}_1, \bar{v}_2, \dots, \bar{v}_n\}$ is a basis for $\mathbb{C}^n$.
- 35. Prove that an $n \times n$ matrix U is unitary if and only if the rows of U form an orthonormal basis for $\mathbb{C}^n$.
- 36. Prove that, if A is a square matrix, then AA^* is a Hermitian matrix.
- 37. Prove that the product of two commuting $n \times n$ Hermitian matrices is also a Hermitian matrix. What can you say about the sum of two Hermitian matrices?
- 38. Prove that the product of two $n \times n$ unitary matrices is also a unitary matrix. What about the sum?
- 39. Let $T: \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a linear transformation whose standard matrix representation is a unitary matrix U . Show that $\langle T(u), T(v) \rangle = \langle u, v \rangle$ for all $u, v \in \mathbb{C}^n$. [HINT: Remember that $\langle u, v \rangle = u^*v$.]
- 40. Prove that for $u, v \in \mathbb{C}^n$, we have $(u^*v)^* = \overline{u^*v} = v^*u = u^T\bar{v}$.
- 41. Describe the unitary diagonal matrices.
- 42. Prove that, if U is unitary, then $\bar{U}$, U^T , and U^* are unitary matrices also.
- 43. A square matrix A is **normal** if $A^*A = AA^*$.
 - a. Show that every Hermitian matrix is normal.
 - b. Show that every unitary matrix is normal.
 - c. Show that, if $A^* = -A$, then A is normal.
- 44. Let A be an $n \times n$ matrix. Referring to Exercise 43, prove that, if A is normal, then $\|Az\| = \|A^*z\|$ for all $z \in \mathbb{C}^n$.
- 45. Prove the converse of the statement in Exercise 44.

MATLAB

MATLAB can work with complex numbers. When typing a complex number $a + bi$ as a component of a vector or as an entry of a matrix, be sure to type $a+b*i$ with the $*$ denoting multiplication and with *no spaces* before or after the $+$ or the $*$. A space before the $+$ would cause MATLAB to interpret $a + b*i$ as two numbers, the real number a followed by another entry containing the complex number bi .

For a matrix A , MATLAB interprets A' as the *conjugate* transpose of A —that is, as the transpose of A with every entry replaced by its conjugate. Here are three more MATLAB functions:

$\text{real}(A)$ is the matrix of real parts of the entries in A ,

$\text{imag}(A)$ is the matrix of complex parts of the entries in A .

$\text{conj}(A)$ is the matrix of conjugates of the entries in A .

Matrices for the exercises that follow are in a file. Some of the exercises ask you to check your answers to some of the more tedious pencil-and-paper computations in the exercises for this section.

- M1. Check Exercise 3. (The file matrices are E3A and E3B.)
- M2. Check Exercise 7.
- M3. Check Exercise 8.
- M4. Check Exercise 9.
- M5. Check Exercise 10.
- M6. Check Exercise 11.
- M7. Check Exercise 12.
- M8. Check Exercise 13.
- M9. Check Exercise 28.
- M10. Check Exercise 29.
- M11. Consider the matrix

$$\begin{bmatrix} 2 - 3i & 3 + 7i & -5 + 2i & 7 - 3i & -10 + 4i \\ 8 - i & 2 + 5i & 11 - 3i & 6 + 2i & 14 - 4i \\ 13 + 2i & 3 - 4i & 9 + 9i & 3 - 2i & 7 + 6i \\ 5 + i & 8 - 4i & 12 + 8i & -3 + 2i & 1 - 5i \end{bmatrix}$$

- a. Find the norms of the four row vectors by multiplying appropriate matrices.
- b. Find the norms of the five column vectors by multiplying appropriate matrices.
- c. Find the inner product $\langle r_1, r_3 \rangle$ where r_1 is the first row vector and r_3 is the third row vector.
- d. Find the inner product $\langle c_2, c_5 \rangle$ where c_2 is the second column vector and c_5 is the fifth column vector.

9.3

EIGENVALUES AND DIAGONALIZATION

Recall the fundamental theorem of real symmetric matrices that we stated without proof as Theorem 6.8:

Every real symmetric matrix is diagonalizable by a real orthogonal matrix.

Our main goal in this section is to extend this result to complex matrices, as follows:

Every Hermitian matrix is diagonalizable by a unitary matrix.

We will prove this theorem, which has the theorem for real symmetric matrices as an easy corollary.

Eigenvalues for Complex Matrices

We begin by extending the notions of eigenvalues and eigenvectors to complex matrices. The definitions are identical to those for real matrices. If A is an $n \times n$ complex matrix and if $Av = \lambda v$, where $\lambda \in \mathbb{C}$ and $v \in \mathbb{C}^n$, $v \neq 0$, then λ is an **eigenvalue** of A and v is a corresponding **eigenvector**. The zero vector and the set of all eigenvectors of A corresponding to λ constitute the **eigenspace** E_λ . Computation of eigenvalues and eigenspaces of a complex matrix is the same as for real matrices, except that the arithmetic involves complex numbers and consequently is more laborious to do with pencil and paper. Every $n \times n$ complex matrix has n not necessarily distinct eigenvalues. This is a consequence of the Fundamental Theorem of Algebra, which we stated in Section 9.1. Recall that, for real matrices, there may exist no real eigenvalues.

EXAMPLE 1 Find the eigenvalues and eigenspaces of the matrix

$$A = \begin{bmatrix} 1 & 0 & i \\ 0 & 2 & 0 \\ -i & 0 & 1 \end{bmatrix}.$$

SOLUTION The characteristic polynomial of A is

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 1 - \lambda & 0 & i \\ 0 & 2 - \lambda & 0 \\ -i & 0 & 1 - \lambda \end{vmatrix} = (2 - \lambda)((1 - \lambda)^2 + i^2) \\ &= (2 - \lambda)(1 - 2\lambda + \lambda^2 - 1) = -\lambda(2 - \lambda)^2. \end{aligned}$$

The three roots of $-\lambda(2 - \lambda)^2 = 0$ are $\lambda_1 = 0$, $\lambda_2 = \lambda_3 = 2$.

For the eigenvalue $\lambda_1 = 0$, we have

$$A - \lambda_1 I = \begin{bmatrix} 1 & 0 & i \\ 0 & 2 & 0 \\ -i & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & i \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

which gives the eigenspace $E_0 = \text{sp} \left(\begin{bmatrix} -i \\ 0 \\ 1 \end{bmatrix} \right)$. For the double root $\lambda_2 = \lambda_3 = 2$, we have

$$A - 2I = \begin{bmatrix} -1 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -i \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

which gives the two-dimensional eigenspace $E_2 = \text{sp} \left(\begin{bmatrix} i \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right)$. ■

Definitions and theorems concerning eigenvalues and eigenvectors that depend only on the field axioms discussed at the beginning of Section 9.2 continue to make sense and to hold for complex matrices. In particular, an $n \times n$ complex matrix A is **diagonalizable** if and only if there exist an invertible matrix C and a diagonal matrix D such that $D = C^{-1}AC$. Just as for real matrices, two complex $n \times n$ matrices A and B are **similar** if there exists an invertible $n \times n$ matrix C such that $B = C^{-1}AC$. Similarity is an equivalence relation. Thus, A is similar to A ; if A is similar to B , then B is similar to A ; and if furthermore B is similar to D , then A is similar to D . All of these things are defined and proved using just field properties.

Consider again the equation $D = C^{-1}AC$, where D is a diagonal matrix. The equivalent equation, $CD = AC$, for an invertible matrix C shows that A is diagonalizable if and only if $\mathbb{C}^n$ has a basis of eigenvectors of A , and it shows that the matrix C must have such a basis of eigenvectors as its column vectors, whereas D has the corresponding eigenvalues on its diagonal. We obtain all of this from $CD = AC$ by considering the j th column vector of CD and comparing it with the j th column vector of AC , just as we did for the real case in Section 5.2. Such a basis for $\mathbb{C}^n$ of eigenvectors of A exists if and only if the algebraic multiplicity of each eigenvalue is equal to its geometric multiplicity (the dimension of the corresponding eigenspace).

EXAMPLE 2 Let $A = \begin{bmatrix} 1 & 0 & i \\ 0 & 2 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Find a matrix C such that $C^{-1}AC$ is a diagonal matrix.

SOLUTION From the preceding example, we see that A has an eigenvalue $\lambda_1 = 0$ of algebraic multiplicity 1 with eigenspace

$$E_0 = \text{sp} \left(\begin{bmatrix} -i \\ 0 \\ 1 \end{bmatrix} \right)$$

and that it has the double eigenvalue $\lambda_2 = \lambda_3 = 2$ with eigenspace

$$E_2 = \text{sp} \left(\begin{bmatrix} i \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right).$$

Thus, the algebraic multiplicity of each eigenvalue is equal to its geometric multiplicity, and the vectors shown form a basis for $\mathbb{C}^3$. Therefore, the matrix

$$C = \begin{bmatrix} -i & i & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

is invertible and diagonalizes A ; and we must also have $C^{-1}AC = D$, where

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}. \quad \blacksquare$$

The proof of Theorem 5.3, which asserts that eigenvectors corresponding to distinct eigenvalues are independent, depends only on field properties and thus is valid in the complex case. Consequently, every matrix having only eigenvalues of algebraic multiplicity 1 is diagonalizable. We focus our attention on the geometric multiplicity of any eigenvalues of algebraic multiplicity greater than 1, when determining whether a matrix is diagonalizable.

EXAMPLE 3 Find all values of c for which the matrix

$$A = \begin{bmatrix} i & c & 1 \\ 0 & i & 2i \\ 0 & 0 & 1 \end{bmatrix}$$

is diagonalizable.

SOLUTION The eigenvalues of the upper-triangular matrix A are $\lambda_1 = \lambda_2 = i$ and $\lambda_3 = 1$. We focus our attention on the eigenvalue i of algebraic multiplicity 2. For A to be diagonalizable, its eigenspace E_i must have geometric multiplicity 2. The eigenspace is the nullspace of the matrix

$$A - \lambda_1 I = \begin{bmatrix} 0 & c & 1 \\ 0 & 0 & 2i \\ 0 & 0 & 1 - i \end{bmatrix},$$

and the nullspace has dimension 2 if and only if the matrix has rank 1, which is the case if and only if $c = 0$. $\blacksquare$

Diagonalization of Hermitian Matrices

Diagonalization via a unitary matrix is of special importance, as we saw in the real case, where it becomes diagonalization by an orthogonal matrix. We call $n \times n$ matrices A and B **unitarily equivalent** if there is a unitary matrix U such

that $B = U^{-1}AU$. Because the inverse of a unitary matrix is unitary and because a product of unitary matrices is unitary, we can show that unitary equivalence is an equivalence relation. Thus, A is unitarily equivalent to itself; and if A is unitarily equivalent to B , then B is unitarily equivalent to A ; if furthermore B is unitarily equivalent to C , then A is unitarily equivalent to C .

Now we achieve the main goal of this section: to prove that Hermitian matrices are unitarily equivalent to a diagonal matrix. That is, a Hermitian matrix can be diagonalized using a unitary matrix. This follows from a very important result known as *Schur's lemma* (or *Schur's unitary triangularization theorem*), which we state, deferring the proof until the end of this section.

THEOREM 9.4 Schur's Lemma

Let A be an $n \times n$ (complex) matrix. There is a unitary matrix U such that $U^{-1}AU$ is upper triangular.

Using Schur's lemma, we can prove that every Hermitian matrix is diagonalizable, and that the diagonalizing matrix can be chosen to be unitary. We express this by saying that every Hermitian matrix is **unitarily diagonalizable**.

THEOREM 9.5 Spectral Theorem for Hermitian Matrices

If A is a Hermitian matrix, there exists a unitary matrix U such that $U^{-1}AU$ is a diagonal matrix. Furthermore, all eigenvalues of A are real.

PROOF By Schur's lemma, there exists a unitary matrix U such that $U^{-1}AU$ is an upper-triangular matrix. Because U is unitary, we have $U^*U = I$, so $U^{-1} = U^*$; and because A is Hermitian, we also know that $A^* = A$. Thus, we have

$$(U^{-1}AU)^* = (U^*AU)^* = U^*A^*(U^*)^* = U^*AU = U^{-1}AU,$$

which shows that the upper-triangular matrix $U^{-1}AU$ is also Hermitian. Because the conjugate transpose of an upper-triangular matrix is a lower-triangular matrix, we see that the entries above the diagonal in $U^{-1}AU$ must all be zero; therefore, $U^{-1}AU = D$, where D is a diagonal matrix. Thus, A is unitarily diagonalizable.

It remains to be shown that each eigenvalue of A is a real number. From the theory of diagonalization, we know that the entries on the diagonal of D are the eigenvalues of A . Now we showed in the preceding paragraph that the matrix $D = U^{-1}AU$ is Hermitian, so $D^* = D$. Forming the conjugate transpose of a diagonal matrix amounts simply to taking the conjugates of the entries on the diagonal. Because $D^* = D$, the entries on the diagonal of D remain unchanged under conjugation, so they must be real numbers. $\blacktriangle$

COROLLARY Fundamental Theorem of Real Symmetric Matrices

Every $n \times n$ real symmetric matrix has n real eigenvalues, counted with their algebraic multiplicity, and is diagonalizable by a real orthogonal matrix.

PROOF Because every real $n \times n$ symmetric matrix A is also Hermitian, Theorem 9.5 establishes that all of its eigenvalues in $\mathbb{C}$ actually lie in $\mathbb{R}$; therefore, the matrix has n real eigenvalues, counting them with their algebraic multiplicity. Furthermore, Theorem 9.5 asserts that A can be diagonalized by a unitary matrix U . We know that the column vectors of U are eigenvectors of A . Now the eigenvectors of A can be computed by row reductions of $A - \lambda_i I$, where the λ_i are eigenvalues of A . Because all the λ_i are real, the row reductions all take place in the field $\mathbb{R}$ of real numbers. The reduced echelon form of $A - \lambda_i I$ is thus a *real* matrix; it must have a nullspace of dimension (geometric multiplicity) equal to the algebraic multiplicity of λ_i , because A is diagonalizable. Thus, we can find bases for the eigenspaces E_{λ_i} consisting of vectors in $\mathbb{R}^n$. Using the Gram–Schmidt process, we can assume that the basis of each eigenspace is orthonormal. The matrix C having as column vectors the vectors in these orthonormal bases of eigenspaces E_{λ_i} is thus a real orthogonal matrix that diagonalizes A . $\blacktriangle$

EXAMPLE 4 Find a unitary matrix that diagonalizes the matrix A in Example 2.

SOLUTION We found in Example 2 that the matrix

$$C = \begin{bmatrix} -i & i & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

diagonalizes A . Notice that the inner product of any two distinct column vectors of this matrix is zero, so the column vectors are orthogonal. We need only normalize them to length 1 in order to obtain a unitary matrix that diagonalizes A . Thus, such a matrix is

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} -i & i & 0 \\ 0 & 0 & \sqrt{2} \\ 1 & 1 & 0 \end{bmatrix}.$$

Recall that in Theorem 6.7 we showed that real eigenvectors of a symmetric matrix corresponding to distinct eigenvalues are orthogonal. Generalizing this, we can show that the eigenvectors of a Hermitian matrix corresponding to distinct eigenvalues are orthogonal. We ask you to show this in Exercise 21, using the fact that a Hermitian matrix can be unitarily diagonalized; but it is easy to demonstrate this orthogonality by using properties of matrices and the fact that the eigenvalues must be real.

THEOREM 9.6 Orthogonality of Eigenspaces of a Hermitian Matrix

The eigenvectors of a Hermitian matrix corresponding to distinct eigenvalues are orthogonal.

PROOF Let $\mathbf{v}$ and $\mathbf{w}$ be eigenvectors of a Hermitian matrix A corresponding to distinct eigenvalues λ_1 and λ_2 , respectively. Using the facts that $A = A^*$ and that the eigenvalues are real, so that $\bar{\lambda}_2 = \lambda_2$, we have

$$\begin{aligned}\lambda_1(\mathbf{w}^*\mathbf{v}) &= \mathbf{w}^*(\lambda_1\mathbf{v}) = \mathbf{w}^*(A\mathbf{v}) = \mathbf{w}^*(A^*\mathbf{v}) = (\mathbf{w}^*A^*)\mathbf{v} \\ &= (A\mathbf{w})^*\mathbf{v} = (\lambda_2\mathbf{w})^*\mathbf{v} = \lambda_2(\mathbf{w}^*\mathbf{v}).\end{aligned}$$

Therefore, $(\lambda_1 - \lambda_2)(\mathbf{w}^*\mathbf{v}) = 0$. Because $\lambda_1 \neq \lambda_2$, we must have $\mathbf{w}^*\mathbf{v} = 0$, so $\mathbf{w}$ and $\mathbf{v}$ are orthogonal. $\blacktriangle$

EXAMPLE 5 Find a unitary matrix C that diagonalizes the Hermitian matrix

$$A = \begin{bmatrix} -1 & i & 1+i \\ -i & 1 & 0 \\ 1-i & 0 & 1 \end{bmatrix}.$$

SOLUTION We find that

$$\begin{aligned}|A - \lambda I| &= \begin{vmatrix} -1-\lambda & i & 1+i \\ -i & 1-\lambda & 0 \\ 1-i & 0 & 1-\lambda \end{vmatrix} \\ &= (-1-\lambda) \begin{vmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} - i \begin{vmatrix} -i & 0 \\ 1-i & 1-\lambda \end{vmatrix} + \\ &\quad (1+i) \begin{vmatrix} -i & 1-\lambda \\ 1-i & 0 \end{vmatrix} \\ &= (-1-\lambda)(1-\lambda)^2 - i(-i)(1-\lambda) + (1+i)(-1)(1-i)(1-\lambda) \\ &= (1-\lambda)(\lambda^2 - 1 - 1 - 2) = (1-\lambda)(\lambda^2 - 4).\end{aligned}$$

Thus, the eigenvalues are $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = -2$. To find U , we need only compute one eigenvector of length 1 for each of these three distinct eigenvalues. The three eigenvectors we obtain must form an orthonormal set, according to Theorem 9.6.

For $\lambda_1 = 1$, we find that

$$A - I = \begin{bmatrix} -2 & i & 1+i \\ -i & 0 & 0 \\ 1-i & 0 & 0 \end{bmatrix},$$

so an eigenvector is

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1+i \\ -i \end{bmatrix}.$$

For $\lambda_2 = 2$, we find that

$$A - 2I = \begin{bmatrix} -3 & i & 1+i \\ -i & -1 & 0 \\ 1-i & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -i & 0 \\ 0 & -2i & 1+i \\ 0 & 1+i & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -i & 0 \\ 0 & 1 & (i-1)/2 \\ 0 & 0 & 0 \end{bmatrix},$$

so a corresponding eigenvector is

$$\mathbf{v}_2 = \begin{bmatrix} 1+i \\ 1-i \\ 2 \end{bmatrix}.$$

Finally, for $\lambda_3 = -2$, we have

$$A + 2I = \begin{bmatrix} 1 & i & 1+i \\ -i & 3 & 0 \\ 1-i & 0 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & i & 1+i \\ 0 & 2 & -1+i \\ 0 & -1-i & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & i & 1+i \\ 0 & 1 & (-1+i)/2 \\ 0 & 0 & 0 \end{bmatrix},$$

and a corresponding eigenvector is

$$\mathbf{v}_3 = \begin{bmatrix} -3-3i \\ 1-i \\ 2 \end{bmatrix}.$$

We normalize the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ and form the column vectors in U from the resulting vectors of magnitude 1, obtaining

$$U = \begin{bmatrix} 0 & (1+i)/(2\sqrt{2}) & (-3-3i)/(2\sqrt{6}) \\ (1+i)/\sqrt{3} & (1-i)/(2\sqrt{2}) & (1-i)/(2\sqrt{6}) \\ -i/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}.$$

A Criterion for Unitary Diagonalization

We have seen that every Hermitian matrix is unitarily diagonalizable, but of course a unitarily diagonalizable matrix need not be Hermitian. For example, the 1×1 matrix $[i]$ is already diagonal, so it is diagonalizable by the identity matrix. However, it is not Hermitian because $[i]^* = [-i]$. There is actually a way to determine whether a square matrix is unitarily diagonalizable, without having to find its eigenvalues and eigenvectors.

DEFINITION 9.5 Normal Matrix

A square matrix A is **normal** if it commutes with its conjugate transpose, so that $AA^* = A^*A$.

Exercises 25 and 26 ask you to prove the following theorem, which gives a criterion for A to be unitarily diagonalizable in terms of matrix multiplication!

THEOREM 9.7 Spectral Theorem for Normal Matrices

A square matrix A is unitarily diagonalizable if and only if it is a normal matrix.

EXAMPLE 6 Determine all values of a such that the matrix

$$A = \begin{bmatrix} i & a \\ 2 & i \end{bmatrix}$$

is unitarily diagonalizable.

SOLUTION In order for the matrix to be unitarily diagonalizable, we must have $AA^* = A^*A$ so that

$$\begin{bmatrix} i & a \\ 2 & i \end{bmatrix} \begin{bmatrix} -i & 2 \\ \bar{a} & -i \end{bmatrix} = \begin{bmatrix} -i & 2 \\ \bar{a} & -i \end{bmatrix} \begin{bmatrix} i & a \\ 2 & i \end{bmatrix}.$$

Equating entries, we obtain

$$\text{row 1, column 1: } 1 + |a|^2 = 1 + 4, \quad \text{so } |a| = 2,$$

$$\text{row 1, column 2: } 2i - ai = -ai + 2i,$$

$$\text{row 2, column 1: } -2i + \bar{a}i = \bar{a}i - 2i,$$

$$\text{row 2, column 2: } 4 + 1 = |a|^2 + 1, \quad \text{so } |a| = 2.$$

Clearly these conditions are satisfied as long as $|a| = 2$, so a can be any number of the form $x + yi$, where $x^2 + y^2 = 4$. ■

Proof of Shur's Lemma

We now prove by induction that, if A is an $n \times n$ matrix, there exists a unitary matrix U such that $U^{-1}AU = U^*AU$ is upper triangular. If $n = 1$, the lemma is trivial. We assume as induction hypothesis that the lemma is true for all matrices of size at most $(n-1) \times (n-1)$, and we proceed to show that it must hold for an $n \times n$ matrix A .

Let λ_1 be an eigenvalue of A , and let $\mathbf{v}_1$ be a corresponding eigenvector of norm 1. We can expand $\{\mathbf{v}_1\}$ to a basis for $\mathbb{C}^n$, and by the Gram-Schmidt process we can transform it into an orthonormal basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. Let U_1 be the unitary matrix whose j th column vector is $\mathbf{v}_j$. Now the first column vector of AU_1 is $A\mathbf{v}_1 = \lambda_1\mathbf{v}_1$. Because the i th row vector of U_1^* is $\mathbf{v}_i^*$, and because the vectors $\mathbf{v}_i$ are mutually orthogonal, we see that the first column vector of the matrix $U_1^*AU_1$ is

$$U_1^*(\lambda_1\mathbf{v}_1) = \begin{bmatrix} \lambda_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

This shows that we can write $U_1^*AU_1$ symbolically as

$$U_1^*AU_1 = \begin{bmatrix} \lambda_1 & \times & \times & \cdots & \times \\ 0 & & & & \\ \vdots & & A_1 & & \\ \vdots & & & & \\ 0 & & & & \end{bmatrix}, \quad (1)$$

where we have denoted the $(n-1) \times (n-1)$ submatrix in the lower right-hand corner of $U_1^*AU_1$ by A_1 . By our induction hypothesis, there exists an $(n-1) \times (n-1)$ unitary matrix C such that $C^*A_1C = B$, where B is upper triangular. Let

$$U_2 = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & & & & \\ \vdots & & C & & \\ \vdots & & & & \\ 0 & & & & \end{bmatrix}, \quad (2)$$

where we have used a symbolic notation similar to that in Eq. (1). Because C is unitary, it is clear that U_2 is a unitary matrix. Now let $U = U_1U_2$. Because $U^*U = U_2^*(U_1^*U_1)U_2 = U_2^*IU_2 = U_2^*U_2 = I$, we see that U is a unitary matrix. Now

$$U^*AU = U_2^*(U_1^*AU_1)U_2. \quad (3)$$

The matrix in parentheses in Eq. (3) is the matrix displayed in Eq. (1). From our definition of U_2 in Eq. (2), we see that the $(n-1) \times (n-1)$ block in the lower right-hand corner of U^*AU is $C^*A_1C = B$. Thus, we have

$$U^*AU = \begin{bmatrix} \lambda_1 & \times & \times & \cdots & \times \\ 0 & & & & \\ \vdots & & B & & \\ \vdots & & & & \\ 0 & & & & \end{bmatrix},$$

which is upper triangular because B is an upper-triangular matrix. This completes our induction argument. $\blacktriangle$

SUMMARY

1. An $n \times n$ matrix A is diagonalizable if and only if $\mathbb{C}^n$ has a basis consisting of eigenvectors of A . Equivalently, each eigenvalue has algebraic multiplicity equal to its geometric multiplicity.
2. Every Hermitian matrix is diagonalizable by a unitary matrix.
3. Every Hermitian matrix has real eigenvalues.

4. A square matrix A is unitarily diagonalizable if and only if it is normal, so that $AA^* = A^*A$.
5. Schur's lemma states that every square matrix is unitarily equivalent to an upper-triangular matrix.

EXERCISES

In Exercises 1–12, find a unitary matrix U and a diagonal matrix D such that $D = U^{-1}AU$ for the given matrix A .

1. $A = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$
2. $A = \begin{bmatrix} 1 & 2i \\ -2i & 1 \end{bmatrix}$
3. $A = \begin{bmatrix} 1 & 1+i \\ 1-i & 2 \end{bmatrix}$
4. $A = \begin{bmatrix} 9 & 3-i \\ 3+i & 0 \end{bmatrix}$
5. $A = \begin{bmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
6. $A = \begin{bmatrix} 1 & -i & 0 \\ i & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$
7. $A = \begin{bmatrix} 1 & 2-2i & 0 \\ 2+2i & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$
8. $A = \begin{bmatrix} 0 & 0 & 1+2i \\ 0 & 5 & 0 \\ 1-2i & 0 & 4 \end{bmatrix}$
9. $A = \begin{bmatrix} 1 & 0 & 2+2i \\ 0 & -3 & 0 \\ 2-2i & 0 & -1 \end{bmatrix}$
10. $A = \begin{bmatrix} 2 & 0 & 1-i \\ 0 & 3 & 0 \\ 1+i & 0 & 1 \end{bmatrix}$
11. $A = \begin{bmatrix} 3 & i & 1+i \\ -i & 1 & 0 \\ 1-i & 0 & 1 \end{bmatrix}$
12. $A = \begin{bmatrix} -3 & 5i & 1+i \\ -5i & 3 & 0 \\ 1-i & 0 & 3 \end{bmatrix}$
13. Find all $a \in \mathbb{C}$ such that the matrix $\begin{bmatrix} i & 4 \\ a & i \end{bmatrix}$ is unitarily diagonalizable.
14. Find all $a, b \in \mathbb{C}$ such that the matrix $\begin{bmatrix} i & a \\ b & i \end{bmatrix}$ is unitarily diagonalizable.
15. Find all $a \in \mathbb{C}$ such that the matrix $\begin{bmatrix} i & a \\ 1 & 3i \end{bmatrix}$ is unitarily diagonalizable.
16. Find all $a, b \in \mathbb{C}$ such that the matrix $\begin{bmatrix} a & -i \\ i & b \end{bmatrix}$ is unitarily diagonalizable.
17. Prove that every 2×2 real matrix that is unitarily diagonalizable has one of the following forms: $\begin{bmatrix} a & b \\ b & d \end{bmatrix}$, $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$, for $a, b, d \in \mathbb{R}$.
18. Determine whether the matrix $\begin{bmatrix} i & -1 & 1 \\ 1 & -i & -1 \\ -1 & 1 & i \end{bmatrix}$ is unitarily diagonalizable.
19. Mark each of the following True or False.
 - a. Every square matrix is unitarily equivalent to a diagonal matrix.
 - b. Every square matrix is unitarily equivalent to an upper-triangular matrix.
 - c. Every Hermitian matrix is unitarily equivalent to a diagonal matrix.
 - d. Every unitarily diagonalizable matrix is Hermitian.
 - e. Every real symmetric matrix is Hermitian.
 - f. Every diagonalizable matrix is normal.
 - g. Every unitarily diagonalizable matrix is normal.
 - h. Every real symmetric matrix is normal.
 - i. Every square matrix is diagonalizable, although perhaps not by a unitary matrix.
 - j. Every square matrix with eigenvalues of algebraic multiplicity 1 is diagonalizable by a unitary matrix.
20. Prove that the eigenvalues of a Hermitian matrix are real, without using Theorem 9.5 or Schur's lemma. [HINT: Let $Av = \lambda v$, and use the fact that $v^*Av = v^*A^*v$.]

21. Argue directly from Theorem 9.5 that eigenvectors from different eigenspaces of a Hermitian matrix are orthogonal.
22. Suppose that A is an $n \times n$ matrix such that $A^* = -A$. Show that
 - a. A has eigenvalues of the form ri , where $r \in \mathbb{R}$,
 - b. A is diagonalizable by a unitary matrix.
 [HINT FOR BOTH PARTS: Work with iA .]
23. Prove that an $n \times n$ matrix A is unitarily diagonalizable if and only if $\|Av\| = \|A^*v\|$ for all $v \in \mathbb{C}^n$.
24. Prove that a normal matrix is Hermitian if and only if all its eigenvalues are in $\mathbb{R}$.
25.
 - a. Prove that a diagonal matrix is normal.
 - b. Prove that, if A is normal and B is unitarily equivalent to A , then B is normal.
 - c. Deduce from parts a and b that a unitarily diagonalizable matrix is normal.
26.
 - a. Prove that every normal matrix A is unitarily equivalent to a normal upper-triangular matrix B . (Use Schur's lemma and part b of Exercise 25.)
 - b. Prove that an $n \times n$ normal upper-triangular matrix B must be diagonal. [HINT: Let $C = B^*B = BB^*$. Equating the computations of c_{11} from B^*B and from BB^* , show that $b_{1j} = 0$ for $1 < j \leq n$. Then equate the computations of c_{22} from B^*B and from BB^* to show that $b_{2j} = 0$ for $2 < j \leq n$. Continue this process to show that B is lower triangular.]
 - c. Deduce from parts a and b that a normal matrix is unitarily diagonalizable.



In Exercises 27 and 28, use the command $[U, D] = \text{eig}(A)$ in MATLAB to work the indicated exercise. If your MATLAB answer U for the unitary matrix differs from the U that we found using pencil and paper and put in the answers at the end of our text, explain how you can get from one answer to the other.

27. Exercise 9

28. Exercise 11

9.4

JORDAN CANONICAL FORM

Jordan Blocks

We have spent considerable time on diagonalization of matrices. The preceding section was concerned primarily with unitary diagonalization. As we have seen, diagonal matrices are easily handled. Unfortunately, not every $n \times n$ matrix A can be diagonalized, because we cannot always find a basis for $\mathbb{C}^n$ consisting of eigenvectors of A . We remind you of this with an example that is well worth studying.

EXAMPLE 1 Show that the matrix

$$J = \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

is not diagonalizable.

SOLUTION We see that 5 is the only eigenvalue of J , and that it has algebraic multiplicity 3. However,

$$J - 5I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

which shows that the eigenspace E_5 has dimension 2 and has basis $\{e_1, e_3\}$. Thus, the geometric multiplicity of the eigenvalue 5 is only 2, and J is not diagonalizable. We cannot find a basis for $\mathbb{R}^3$ (or even $\mathbb{C}^3$) consisting entirely of eigenvectors of J . ■

Let us examine the matrix J in Example 1 a bit more. Notice that, although $J - 5I$ has a nullspace of dimension 2, the matrix $(J - 5I)^2$ is the zero matrix and has all of $\mathbb{C}^3$ as nullspace. Moreover, multiplication on the left by $J - 5I$ carries e_2 into e_1 and carries both e_1 and e_3 into 0 . We say that $J - 5I$ **annihilates** e_1 and e_3 . The action of $J - 5I$ on these standard basis vectors is denoted schematically by the two *strings*

$$J - 5I: \quad \begin{array}{l} e_2 \rightarrow e_1 \rightarrow 0, \\ e_3 \rightarrow 0. \end{array} \quad (1)$$

Diagram (1) also shows that $(J - 5I)^2$ maps each of these basis vectors into 0 . Because $(J - 5I)e_2 = e_1$, we have $Je_2 = 5e_2 + e_1$, whereas $Je_1 = 5e_1$ and $Je_3 = 5e_3$.

EXAMPLE 2 Let

$$J = \begin{bmatrix} \lambda & 1 & 0 & 0 & 0 \\ 0 & \lambda & 1 & 0 & 0 \\ 0 & 0 & \lambda & 1 & 0 \\ 0 & 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & 0 & \lambda \end{bmatrix}. \quad (2)$$

Discuss the action of $J - \lambda I$ on the standard basis vectors, drawing a schematic diagram similar to diagram (1). Describe also the action of J on the vectors in the standard basis.

SOLUTION We find that

$$J - \lambda I = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

We see that multiplication of e_i on the left by $J - \lambda I$ yields e_{i-1} for $2 \leq i \leq 5$, whereas $(J - \lambda I)e_1 = 0$. Schematically, we have just one *string*,

$$J - \lambda I: \quad e_5 \rightarrow e_4 \rightarrow e_3 \rightarrow e_2 \rightarrow e_1 \rightarrow 0. \quad (3)$$

Left-multiplication by J yields

$$\begin{aligned} Je_5 &= \lambda e_5 + e_4, & Je_4 &= \lambda e_4 + e_3, & Je_3 &= \lambda e_3 + e_2, \\ Je_2 &= \lambda e_2 + e_1, & Je_1 &= 0. \end{aligned}$$

The matrix J in Example 2 is an example of a *Jordan block* matrix.

DEFINITION 9.6 Jordan Block

An $m \times m$ matrix is a **Jordan block** if it is structured as follows:

1. All diagonal entries are equal.
2. Each entry immediately above a diagonal entry is 1.
3. All other entries are zero.

Thus, the matrix J in Example 2 is a Jordan block. However, the matrix in Example 1 is not a Jordan block, since the entry 5 at the bottom of the diagonal does not have a 1 just above it. A Jordan block has the properties described in the next theorem. These properties were illustrated in Example 2, and we leave a formal proof to you if you desire one. Notice that, for an $m \times m$ Jordan block

$$J = \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ & & & \ddots & & \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{bmatrix},$$

we have just one string:

$$J - \lambda I: \quad e_m \rightarrow e_{m-1} \rightarrow \cdots \rightarrow e_2 \rightarrow e_1 \rightarrow 0.$$

THEOREM 9.8 Properties of a Jordan Block

Let J be an $m \times m$ Jordan block with diagonal entries all equal to λ . Then the following properties hold:

1. $(J - \lambda I)e_i = e_{i-1}$ for $1 < i \leq m$, and $(J - \lambda I)e_1 = 0$.
2. $(J - \lambda I)^m = 0$, but $(J - \lambda I)^i \neq 0$ for $i < m$.
3. $Je_i = \lambda e_i + e_{i-1}$ for $1 < i \leq m$, whereas $Je_1 = \lambda e_1$.

Jordan Canonical Forms

We have seen that not every $n \times n$ matrix is diagonalizable. It is our purpose in this section to show that every $n \times n$ matrix is similar to a matrix having all

entries 0 except for those on the diagonal and entries 1 immediately above some diagonal entries; each 1 above a diagonal entry must have the same number on its left as below it on the diagonal. An example of such a matrix is

$$J = \begin{bmatrix} -i & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -i & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 5 \end{bmatrix}. \quad (4)$$

As the shading indicates, this matrix J is comprised of four Jordan blocks, placed corner-to-corner along the diagonal.

DEFINITION 9.7 Jordan Canonical Form

An $n \times n$ matrix J is a **Jordan canonical form** if it consists of Jordan blocks, placed corner-to-corner along the main diagonal, as in matrix (4), with only zero entries outside these Jordan blocks.

Every diagonal matrix is a Jordan canonical form, because each diagonal entry can be viewed as being the sole entry in a 1×1 Jordan block. Notice that matrix (4) contains the 1×1 Jordan block [2]. Notice, too, that the breaks between the Jordan blocks in matrix (4) occur where some diagonal entry has a 0 rather than 1 immediately above it.

EXAMPLE 3 Is the matrix

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 7 \end{bmatrix}$$

a Jordan canonical form? Why?

SOLUTION This matrix is not a Jordan canonical form. Because not all diagonal entries are equal, there should be at least two Jordan blocks present in order for the matrix to be a Jordan canonical form, and [7] should be a 1×1 Jordan block. However, the entry immediately above 7 is not 0. Consequently, this matrix is not a Jordan canonical form. ■

EXAMPLE 4 Describe the effect of matrix J in Eq. (4) on each of the standard basis vectors in $\mathbb{C}^8$. Then give the eigenvalues and eigenspaces of J . Finally, find the dimension of the nullspace of $(J - \lambda I)^k$ for each eigenvalue λ of J and for each positive integer k .

SOLUTION We find that

$$\begin{aligned}Je_3 &= -ie_3 + e_3, & Je_2 &= -ie_2 + e_1, & Je_1 &= -ie_1, \\Je_5 &= -ie_5 + e_4, & Je_4 &= -ie_4, \\Je_6 &= 2e_6, \\Je_8 &= 5e_8 + e_7, & Je_7 &= 5e_7.\end{aligned}$$

The eigenvalues of J are $-i$, 2 , and 5 , which have algebraic multiplicities of 5 , 1 , and 2 , respectively. The eigenspaces of J are $E_{-i} = \text{sp}(e_1, e_4)$, $E_2 = \text{sp}(e_6)$, and $E_5 = \text{sp}(e_7)$, as you can easily check.

The effect of $J - (-i)I$ on the first five standard basis vectors is given by the two strings

$$J + iI: \quad \begin{array}{l} e_3 \rightarrow e_2 \rightarrow e_1 \rightarrow 0, \\ e_5 \rightarrow e_4 \rightarrow 0. \end{array} \quad (5)$$

The 3×3 lower right-hand corner of $J + iI$ describes the action of $J + iI$ on e_6 , e_7 , and e_8 . Because this 3×3 matrix has a nonzero determinant, it causes $J + iI$ to carry these three vectors into three independent vectors, and the same is true of all powers of $J + iI$. Thus we can determine the dimension of the nullspace of $J + iI$ by diagram (5), and we find that

- $J + iI$ has nullspace $\text{sp}(e_1, e_4)$ of dimension 2,
- $(J + iI)^2$ has nullspace $\text{sp}(e_1, e_2, e_4, e_5)$ of dimension 4,
- $(J + iI)^3$ has nullspace $\text{sp}(e_1, e_2, e_3, e_4, e_5)$ of dimension 5,
- $(J + iI)^k$ has the same nullspace as that of $(J + iI)^3$ for $k > 3$.

By a similar argument, we find that

- $(J - 2I)^k$ has nullspace $\text{sp}(e_6)$ of dimension 1 for $k \geq 1$,
- $J - 5I$ has nullspace $\text{sp}(e_7)$ of dimension 1,
- $(J - 5I)^k$ has nullspace $\text{sp}(e_7, e_8)$ of dimension 2 for $k > 1$. ■

HISTORICAL NOTE THE JORDAN CANONICAL FORM appears in the *Treatise on Substitutions and Algebraic Equations*, the chief work of the French algebraist Camille Jordan (1838–1921). This text, which appeared in 1870, incorporated the author's group-theory work over the preceding decade and became the bible of the field for the remainder of the nineteenth century. The theorem containing the canonical form actually deals not with matrices over the real numbers, but with matrices with entries from the finite field of order p . And as the title of the book indicates, Jordan was not considering matrices as such, but the linear substitutions that they represented.

Camille Jordan, a brilliant student, entered the Ecole Polytechnique in Paris at the age of 17 and practiced engineering from the time of his graduation until 1885. He thus had ample time for mathematical research. From 1873 until 1912, he taught at both the Ecole Polytechnique and the Collège de France. Besides doing seminal work on group theory, he is known for important discoveries in modern analysis and topology.

EXAMPLE 5 Suppose a 9×9 Jordan canonical form J has the following properties:

1. $(J - 3iI)^k$ has rank 7 for $k = 1$, rank 5 for $k = 2$, and rank 4 for $k \geq 3$.
2. $(J + I)^j$ has rank 6 for $j = 1$ and rank 5 for $j \geq 2$.

Find the Jordan blocks that appear in J .

SOLUTION Because the rank of $J - 3iI$ is 7, the dimension of its nullspace is $9 - 7 = 2$, so $3i$ is an eigenvalue of geometric multiplicity 2. It must give rise to two Jordan blocks. In addition, $J - 3iI$ must annihilate two eigenvectors $\mathbf{e}_r$ and $\mathbf{e}_s$ in the standard basis. Because the rank of $(J - 3iI)^2$ is 5, its nullspace must have dimension 4, so in a diagram of the effect of $J - 3iI$ on the standard basis, we must have $(J - 3iI)\mathbf{e}_{r+1} = \mathbf{e}_r$ and $(J - 3iI)\mathbf{e}_{s+1} = \mathbf{e}_s$. Because $(J - 3iI)^k$ has rank 4 for $k \geq 3$, its nullity is 5, and we have just one more standard basis vector—either $\mathbf{e}_{r+2}$ or $\mathbf{e}_{s+2}$ —that is annihilated by $(J - 3iI)^3$. Thus, the two Jordan blocks in J that have $3i$ on the diagonal are

$$J_1 = \begin{bmatrix} 3i & 1 & 0 \\ 0 & 3i & 1 \\ 0 & 0 & 3i \end{bmatrix} \quad \text{and} \quad J_2 = \begin{bmatrix} 3i & 1 \\ 0 & 3i \end{bmatrix}.$$

Because $J + I$ has rank 6, its nullspace has dimension $9 - 6 = 3$, so -1 is an eigenvalue of geometric multiplicity 3 and gives rise to three Jordan blocks. Because $(J + I)^j$ has rank 5 for $j \geq 2$, its nullspace has dimension 4, so $(J + I)^2$ annihilates a total of four standard basis vectors. Thus, just one of these Jordan blocks is 2×2 , and the other two are 1×1 . The Jordan blocks arising from the eigenvalue -1 are then

$$J_3 = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix} \quad \text{and} \quad J_4 = J_5 = [-1].$$

The matrix J might have these blocks in any order down its diagonal. Symbolically, we might have

$$J = \begin{bmatrix} J_3 & & 0 \\ & J_1 & \\ 0 & & J_2 \\ & & & J_4 \\ & & & & J_5 \end{bmatrix}, \quad J = \begin{bmatrix} J_4 & & 0 \\ & J_5 & \\ 0 & & J_2 \\ & & & J_1 \\ & & & & J_3 \end{bmatrix},$$

or any other order. ■

Jordan Bases

If an $n \times n$ matrix A is similar to a Jordan canonical form J , we call J a **Jordan canonical form of A** . When this is the case, there exists an invertible matrix C such that $C^{-1}AC = J$. We know that similar matrices represent the same linear

transformation, but with respect to different bases. Thus, if A is similar to J , there must exist a basis $\{b_1, b_2, \dots, b_n\}$ of $\mathbb{C}^n$ with the same schematic string properties relative to A that the standard ordered basis has relative to the matrix J . We proceed to define such a *Jordan basis*.

DEFINITION 9.8 Jordan Basis

Let A be an $n \times n$ matrix. An ordered basis $B = (b_1, b_2, \dots, b_n)$ of $\mathbb{C}^n$ is a **Jordan basis for A** if, for $1 \leq j \leq n$, we have either $Ab_j = \lambda b_j$ or $Ab_j = \lambda b_j + b_{j-1}$, where λ is an eigenvalue of A that we say is **associated with b_j** . If $Ab_j = \lambda b_j + b_{j-1}$, we require that the eigenvalue associated with b_{j-1} also be λ .

If an $n \times n$ matrix A has a Jordan basis B , then the matrix representation of the linear transformation $T(z) = Az$ relative to B must be a Jordan canonical form. We know then that $J = C^{-1}AC$, where C is the $n \times n$ matrix whose j th column vector is the j th vector b_j in B . In a moment we will prove that, for every square matrix, there is an associated Jordan basis, and consequently that every square matrix is similar to a Jordan canonical form. First, though, we outline a method for the computation of a Jordan canonical form of A .

Finding a Jordan canonical form of A

1. Find the eigenvalues of A .
2. For each eigenvalue λ , compute the rank of $(A - \lambda I)$ for consecutive values of k starting with $k = 1$, until the same rank is obtained for two consecutive values of k .
3. From the data generated, find a Jordan canonical form for A , as in Example 5.

We now illustrate this technique.

EXAMPLE 6 Find a Jordan canonical form of the matrix

$$A = \begin{bmatrix} 2 & 5 & 0 & 0 & 1 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}.$$

SOLUTION Because A is an upper-triangular matrix, we see that the eigenvalues of A are $\lambda_1 = \lambda_2 = 2$ and $\lambda_3 = \lambda_4 = \lambda_5 = -1$. Now

$$A - \lambda_1 I = A - 2I = \begin{bmatrix} 0 & 5 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & -3 \end{bmatrix}$$

has rank 4 and consequently has a nullspace of dimension 1. We find that

$$(A - 2I)^2 = \begin{bmatrix} 0 & 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 0 & 9 \end{bmatrix},$$

which has rank 3 and therefore has a nullspace of dimension 2. Furthermore,

$$(A - 2I)^3 = \begin{bmatrix} 0 & 0 & 0 & 0 & 9 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -27 & 0 & 0 \\ 0 & 0 & 0 & -27 & 0 \\ 0 & 0 & 0 & 0 & -27 \end{bmatrix}$$

has the same rank and nullity as $(A - 2I)^2$. Thus we have $Ab_1 = 2b_1$ and $Ab_2 = 2b_2 + b_1$ for some Jordan basis $B = (b_1, b_2, b_3, b_4, b_5)$ for A . There is just one Jordan block associated with $\lambda_1 = 2$ —namely,

$$J_1 = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

For the eigenvalue $\lambda_3 = -1$, we find that

$$A - \lambda_3 I = A + I = \begin{bmatrix} 3 & 5 & 0 & 0 & 1 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

which has rank 2 and therefore has a nullspace of dimension 3. Because -1 is an eigenvalue of both algebraic multiplicity and geometric multiplicity 3, we realize that $J_2 = J_3 = J_4 = [-1]$ are the remaining Jordan blocks. This is confirmed by the fact that

$$(A + I)^2 = \begin{bmatrix} 9 & 30 & 0 & 0 & 3 \\ 0 & 9 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

again has rank 2 and nullity 3. Thus, a Jordan canonical form for A is

$$J = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}.$$

EXAMPLE 7 Find a Jordan basis for matrix A in Example 6.

SOLUTION For the part of a Jordan basis associated with the eigenvalue 2, we need to find a vector $\mathbf{b}_2$ in the nullspace of $(A - 2I)^2$ that is not in the nullspace of $A - 2I$; then we may take $\mathbf{b}_1 = (A - 2I)\mathbf{b}_2$. From the computation of $A - 2I$ and $(A - 2I)^2$ in Example 6, we see that we can take

$$\mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \text{and then} \quad \mathbf{b}_1 = (A - 2I)\mathbf{b}_2 = \begin{bmatrix} 5 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

For $\mathbf{b}_3$, $\mathbf{b}_4$, and $\mathbf{b}_5$, we need only take a basis for the nullspace of $A + I$. We see that we can take

$$\mathbf{b}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{b}_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{b}_5 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 3 \end{bmatrix}.$$

In Example 7, it was easy to find vectors in a Jordan basis corresponding to the eigenvalue 2 whose geometric multiplicity is less than its algebraic multiplicity, because only one Jordan block corresponds to the eigenvalue 2. We now indicate how a Jordan basis can be constructed when more than one such block corresponds to a single eigenvalue λ . Let N_r be the nullspace of $(A - \lambda I)^r$ for $r \geq 1$, and suppose (for example) that $\dim(N_1) = 4$, $\dim(N_2) = 7$, and $\dim(N_3) = 8$ for $r \geq 3$. Then a Jordan basis for A contains four strings corresponding to λ , which we may represent as

$$\mathbf{b}_3 \rightarrow \mathbf{b}_2 \rightarrow \mathbf{b}_1 \rightarrow 0,$$

$$\mathbf{b}_5 \rightarrow \mathbf{b}_4 \rightarrow 0,$$

$$\mathbf{b}_7 \rightarrow \mathbf{b}_6 \rightarrow 0,$$

$$\mathbf{b}_8 \rightarrow 0.$$

To find the first and longest of these strings, we compute a basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_8\}$ for the nullspace N_3 of $(A - \lambda I)^3$. The preceding strings show that multiplication of all of the vectors in N_3 on the left by $(A - \lambda I)^2$ yields a space of dimension 1, so at least one of the vectors $\mathbf{v}_i$ has the property that $(A - \lambda I)^2 \mathbf{v}_i \neq 0$.

Let $\mathbf{b}_3$ be such a vector, and set $\mathbf{b}_2 = (A - \lambda I)\mathbf{b}_3$ and $\mathbf{b}_1 = (A - \lambda I)\mathbf{b}_2$. It is not difficult to show that $\mathbf{b}_1$, $\mathbf{b}_2$, and $\mathbf{b}_3$ must be independent. Thus we have found the first string.

Now $\mathbf{b}_1$ and $\mathbf{b}_2$ lie in N_2 , and we can expand the independent set $\{\mathbf{b}_1, \mathbf{b}_2\}$ to a basis $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{w}_1, \dots, \mathbf{w}_3\}$ of N_2 . Again, the strings displayed earlier show that multiplication of the vectors in N_2 on the left by $A - \lambda I$ must yield a space of dimension 3, so there exist two vectors $\mathbf{w}_i$ and $\mathbf{w}_j$ such that the vectors $\mathbf{b}_1$, $(A - \lambda I)\mathbf{w}_i$, and $(A - \lambda I)\mathbf{w}_j$ are independent. Let $\mathbf{b}_5 = \mathbf{w}_i$ and $\mathbf{b}_4 = (A - \lambda I)\mathbf{b}_5$, while $\mathbf{b}_7 = \mathbf{w}_j$ and $\mathbf{b}_6 = (A - \lambda I)\mathbf{b}_7$. It can be shown that the vectors $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_7$ are independent. Finally, we expand the set $\{\mathbf{b}_1, \mathbf{b}_4, \mathbf{b}_6\}$ to a basis $\{\mathbf{b}_1, \mathbf{b}_4, \mathbf{b}_6, \mathbf{b}_8\}$ for N_1 to complete the portion of the Jordan basis corresponding to λ .

Although we know the techniques for finding bases for the nullspaces N_i and for expanding a given set of independent vectors to a basis, significant pencil-and-paper illustrations of this construction would be cumbersome, so we do not include them here. Any Jordan bases requested in the exercises can be found as in Example 7.

An application of the Jordan canonical form to differential equations is indicated in Exercise 32. We mention that computer-aided computation of a Jordan canonical form for a square matrix is not a stable process. Consider, for example, the matrix

$$A = \begin{bmatrix} 2 & c \\ 0 & 2 \end{bmatrix}.$$

If $c = 10^{-100}$, then the Jordan canonical form of A has 1 as its entry in the upper right-hand corner; but if $c = 0$, that entry is 0.

Existence of a Jordan Form for a Square Matrix

To demonstrate the existence of a Jordan canonical form similar to an $n \times n$ matrix A , we need only show that we have a Jordan basis B for A . Let us formalize the concept of a *string* in a Jordan basis $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$. Let λ be an eigenvalue of A . If $A\mathbf{b}_i = \lambda\mathbf{b}_i$ and $A\mathbf{b}_k = \lambda\mathbf{b}_k + \mathbf{b}_{k-1}$ for $i < k < j$, while $A\mathbf{b}_j \neq \lambda\mathbf{b}_j + \mathbf{b}_{j-1}$, we refer to the sequence $\mathbf{b}_i, \mathbf{b}_{i+1}, \dots, \mathbf{b}_{j-1}$ as a *string* of basis vectors **starting at $\mathbf{b}_i$, ending at $\mathbf{b}_{j-1}$, and associated with λ** . This string is represented by the diagram

$$A - \lambda I: \quad \mathbf{b}_{j-1} \rightarrow \dots \rightarrow \mathbf{b}_{i+1} \rightarrow \mathbf{b}_i \rightarrow 0.$$

THEOREM 9.9 Jordan Canonical Form of a Square Matrix

Let A be a square matrix. There exists an invertible matrix C such that the matrix $J = C^{-1}AC$ is a Jordan canonical form. This Jordan canonical form is unique, except for the order of the Jordan blocks of which it is composed.

PROOF We use a proof due to Filippov. First we note that it suffices to prove the theorem for matrices A having 0 as an eigenvalue. Observe that, if λ is an eigenvalue of A , then 0 is an eigenvalue of $A - \lambda I$. Now if we can find C such that $C^{-1}(A - \lambda I)C = J$ is a Jordan canonical form, then $C^{-1}AC = J + \lambda I$ is also a Jordan canonical form. Thus, we restrict ourselves to the case where A has an eigenvalue of 0.

In order to find a Jordan canonical form for A , it is useful to consider also the linear transformation $T: \mathbb{C}^n \rightarrow \mathbb{C}^n$, where $T(z) = Az$; a Jordan basis for A is considered to be a Jordan basis for T . We will prove the existence of a Jordan basis for any such linear transformation by induction on the dimension of the domain of the transformation.

If T is a linear transformation of a one-dimensional vector space $\text{sp}(z)$, then $T(z) = \lambda z$ for some $\lambda \in \mathbb{C}$, and $\{z\}$ is the required Jordan basis. (The matrix of T with respect to this ordered basis is the 1×1 matrix $[\lambda]$, which is already a Jordan canonical form.)

Now suppose that there exist Jordan bases for linear transformations on subspaces of $\mathbb{C}^n$ of dimension less than n , and let $T(z) = Az$ for $z \in \mathbb{C}^n$ and an $n \times n$ matrix A . As noted, we can assume that zero is an eigenvalue of A . Then $\text{rank}(A) < n$; let $r = \text{rank}(A)$. Now T maps $\mathbb{C}^n$ onto the column space of A that is of dimension $r < n$. Let T' be the induced linear transformation of the column space of A into itself, defined by $T'(v) = T(v)$ for v in the column space of A . By our induction hypothesis, there is a Jordan basis

$$B' = (u_1, u_2, \dots, u_r)$$

for this column space of A .

Let S be the intersection of the column space and the nullspace of A . We wish to separate the vectors in B' that are in S from those that are not. The nonzero vectors in S are precisely the eigenvectors in the column space of A with corresponding eigenvalue 0; that is, they are the eigenvectors of T' with eigenvalue 0. In other words, S is the nullspace of T' . Let J' be the matrix representation of T' relative to B' . Because J' is a Jordan canonical form, we see that the nullity of T' (and of J') is precisely the number of zero rows in J' . This is true because J' is an upper-triangular square matrix; it can be brought to echelon form by means of row exchanges that place the zero rows at the bottom while sliding the nonzero rows up. Thus, if $\dim(S) = s$, there are s zero rows in J' . Now in J' we have exactly one zero row for each Jordan block corresponding to the eigenvalue 0—namely, the row containing the bottom row of the block. Because the number of such blocks is equal to the number of strings in B' ending in S , we conclude that there are s such strings. Some of these strings may be of length 1 whereas others may be longer.

Figure 9.11 shows one possible situation when $s = 2$, where two vectors in S —namely, u_1 and u_4 —are ending points of strings

$$u_3 \rightarrow u_2 \rightarrow u_1 \rightarrow 0 \quad \text{and} \quad u_5 \rightarrow u_4 \rightarrow 0$$

lying in the column space of A . These s strings of B' that end in S start at s vectors in the column space of A ; these are the vectors u_3 and u_5 in Figure 9.11.

Because the vector at the beginning of the j th string is in the column space of A , it must have the form $A\mathbf{w}_j$ for some vector $\mathbf{w}_j$ in $\mathbb{C}^n$. Thus we obtain the vectors $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_s$ illustrated in Figure 9.11 for $s = 2$.

Finally, the nullspace of A has dimension $n - r$, and we can expand the set of s independent vectors in S to a basis for this nullspace. This gives rise to $n - r - s$ more vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n-r-s}$. Of course, each $\mathbf{v}_i$ is an eigenvector with corresponding eigenvalue 0.

We claim that

$$(\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{w}_1, \dots, \mathbf{w}_s, \mathbf{v}_1, \dots, \mathbf{v}_{n-r-s})$$

can be reordered to become a Jordan basis B for A (and of course for T). We reorder it by moving the vectors $\mathbf{w}_j$, tucking each one in so that it starts the appropriate string in B' that was used to define it. For the situation in Figure 9.11, we obtain

$$(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{w}_1, \mathbf{u}_4, \mathbf{u}_5, \mathbf{w}_2, \mathbf{u}_6, \dots, \mathbf{u}_r, \mathbf{v}_1, \dots, \mathbf{v}_{n-r-2})$$

as Jordan basis. From our construction, we see that B is a Jordan basis for A if it is a basis for $\mathbb{C}^n$. Because there are $r + s + (n - r - s) = n$ vectors in all, we need only show that they are independent.

Suppose that

$$\sum_{i=1}^r a_i \mathbf{u}_i + \sum_{j=1}^s c_j \mathbf{w}_j + \sum_{k=1}^{n-r-s} d_k \mathbf{v}_k = \mathbf{0}. \quad (6)$$

Because the vectors $\mathbf{v}_k$ lie in the nullspace of A , if we apply A to both sides of this equation, we obtain

$$\sum_{i=1}^r a_i A\mathbf{u}_i + \sum_{j=1}^s c_j A\mathbf{w}_j = \mathbf{0}. \quad (7)$$

Because each $A\mathbf{u}_i$ is either of the form $\lambda \mathbf{u}_i$ or of the form $\lambda \mathbf{u}_i + \mathbf{u}_{i-1}$, we see that the first sum is a linear combination of vectors $\mathbf{u}_i$. Moreover, these vectors

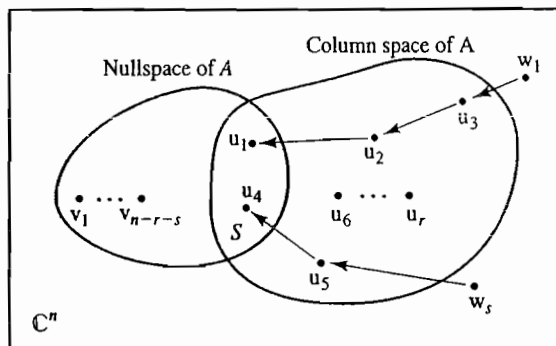


FIGURE 9.11
Construction of a Jordan basis for A ($s = 2$).

Au_i do not begin any string in B' . Now the vectors Aw_i in the second sum are vectors u_i that appear at the start of the s strings in B' that end in S . Thus they do not appear in the first sum. Because B' is an independent set, all the coefficients c_j in Eq. (7) must be zero. Equation (6) can then be written as

$$\sum_{i=1}^r a_i u_i = \sum_{k=1}^{n-r-s} -d_k v_k. \quad (8)$$

Now the vector on the left-hand side of this equation lies in the column space of A , whereas the vector on the right-hand side is in the nullspace of A . Consequently, this vector lies in S and is a linear combination of the s basis vectors u_i in S . Because the v_k were obtained by extending these s vectors to a basis for the nullspace of A , the vector $\mathbf{0}$ is the only linear combination of the v_k that lies in S . Thus, the vector on both sides of Eq. (8) is $\mathbf{0}$. Because the v_k are independent, we see that all d_k are zero. Because the u_i are independent, it follows that the a_i are all zero. Therefore, B is an independent set of n vectors and is thus a basis for $\mathbb{C}^n$. We have seen that, by our construction, it must be a Jordan basis. This completes the induction part of our proof, demonstrating the existence of a Jordan canonical form for every square matrix A .

Our work prior to this theorem makes clear that the Jordan blocks constituting a Jordan canonical form for A are completely determined by the ranks of the matrices $(A - \lambda I)^k$ for all eigenvalues λ of A and for all positive integers k . Thus, a Jordan canonical form J for A is unique except as to the order in which these blocks appear along the diagonal of J . $\blacktriangle$

SUMMARY

1. A Jordan block is a square matrix with all diagonal entries equal, all entries immediately above diagonal entries equal to 1, and all other entries equal to 0.
2. Properties of a Jordan block are given in Theorem 9.8.
3. A square matrix is a Jordan canonical form if it consists of Jordan blocks placed corner to corner along its main diagonal, with entries elsewhere equal to 0.
4. A Jordan basis (see Definition 9.8) for an $n \times n$ matrix A gives rise to a Jordan canonical form J that is similar to A .
5. A Jordan canonical form similar to an $n \times n$ matrix A can be computed if we know the eigenvalues λ_i of A and if we know the rank of $(A - \lambda_i I)^k$ for each λ_i and for all positive integers k .
6. Every square matrix has a Jordan canonical form; that is, it is similar to a Jordan canonical form.

EXERCISES

In Exercises 1–6, determine whether the given matrix is a Jordan canonical form.

1.
$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

2.
$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}$$

3.
$$\begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

4.
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

5.
$$\begin{bmatrix} i & 1 & 0 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

6.
$$\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

In Exercises 7–10:

- Find the eigenvalues of the given matrix J .
- Give the rank and nullity of $(J - \lambda)^k$ for each eigenvalue λ of J and for every positive integer k .
- Draw schemata of the strings of vectors in the standard basis arising from the Jordan blocks in J .
- For each standard basis vector e_k , express Je_k as a linear combination of vectors in the standard basis.

7.
$$\begin{bmatrix} -2 & 1 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

8.
$$\begin{bmatrix} i & 0 & 0 & 0 & 0 \\ 0 & i & 1 & 0 & 0 \\ 0 & 0 & i & 0 & 0 \\ 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & -2 \end{bmatrix}$$

9.
$$\begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

10.
$$\begin{bmatrix} i & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & i & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & i & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & i & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

In Exercises 11–14, find a Jordan canonical form for A from the given data.

- A is 5×5 , $A - 3I$ has nullity 2, $(A - 3I)^2$ has nullity 3, $(A - 3I)^3$ has nullity 4, $(A - 3I)^k$ has nullity 5 for $k \geq 4$.
- A is 7×7 , $A + I$ has nullity 3, $(A + I)^k$ has nullity 5 for $k \geq 2$; $A + iI$ has nullity 1, $(A + iI)^j$ has nullity 2 for $j \geq 2$.
- A is 8×8 , $A - I$ has nullity 2, $(A - I)^2$ has nullity 4, $(A - I)^k$ has nullity 5 for $k \geq 3$; $(A + 2I)^j$ has nullity 3 for $j \geq 1$.
- A is 8×8 ; $A + iI$ has rank 4, $(A + iI)^2$ has rank 2, $(A + iI)^3$ has rank 1, $(A + iI)^k = O$ for $k \geq 4$.

In Exercises 15–22, find a Jordan canonical form and a Jordan basis for the given matrix.

15.
$$\begin{bmatrix} -10 & 4 \\ -25 & 10 \end{bmatrix}$$

16.
$$\begin{bmatrix} 5 & -4 \\ 9 & -7 \end{bmatrix}$$

17.
$$\begin{bmatrix} 4 & 0 & 0 \\ 2 & 1 & 3 \\ 5 & 0 & 4 \end{bmatrix}$$

18.
$$\begin{bmatrix} -3 & 0 & 1 \\ 2 & -2 & 1 \\ -1 & 0 & -1 \end{bmatrix}$$

19.
$$\begin{bmatrix} 2 & 5 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

20.
$$\begin{bmatrix} i & 0 & 0 & 0 & 0 \\ 0 & i & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 2 & 0 & -1 & 0 & 2 \end{bmatrix}$$

$$21. \begin{bmatrix} 2 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

$$22. \begin{bmatrix} 1 & 2 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

23. Mark each of the following True or False.

- ___ a. Every Jordan block matrix has just one eigenvalue.
- ___ b. Every matrix having a unique eigenvalue is a Jordan block.
- ___ c. Every diagonal matrix is a Jordan canonical form.
- ___ d. Every square matrix is similar to a Jordan canonical form.
- ___ e. Every square matrix is similar to a unique Jordan canonical form.
- ___ f. Every 1×1 matrix is similar to a unique Jordan canonical form.
- ___ g. There is a Jordan basis for every square matrix A .
- ___ h. There is a unique Jordan basis for every square matrix A .
- ___ i. Every 3×3 diagonalizable matrix is similar to exactly six Jordan canonical forms.
- ___ j. Every 3×3 matrix is similar to exactly six Jordan canonical forms.

$$24. \text{ Let } A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \text{ Compute}$$

A^2 , A^3 , and A^4 .

25. Let A be an $n \times n$ upper-triangular matrix with all diagonal entries 0. Compute A^m for all positive integers $m \geq n$. (See Exercise 24.) Prove that your answer is correct.

$$26. \text{ Let } A = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}. \text{ Compute}$$

$(A - 2I)^3(A - 3I)^3$.

$$27. \text{ Let } A = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & i \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}. \text{ Compute}$$

$(A - 2I)^2(A - 3I)^3$. Compare with Exercise 26.

$$28. \text{ Let } A = \begin{bmatrix} i & 0 & 0 & 0 & 0 \\ 0 & i & 1 & 0 & 0 \\ 0 & 0 & i & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}. \text{ Find a polynomial in}$$

A (that is, a sum of terms $a_j A^j$ with a term $a_0 I$) that gives the zero matrix. (See Exercises 24–27.)

29. Repeat Exercise 28 for the matrix $A =$

$$\begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & i & 0 \\ 0 & 0 & 0 & 0 & i \end{bmatrix}.$$

30. The Cayley–Hamilton theorem states that, if $p(\lambda) = a_n \lambda^n + \cdots + a_1 \lambda + a_0$ is the characteristic polynomial of a matrix A , then $p(A) = a_n A^n + \cdots + a_1 A + a_0 I = O$, the zero matrix. Prove it. [HINT: Consider $(A - \lambda_i I)^{n_i} \mathbf{b}$, where $\mathbf{b}$ is a vector in a Jordan basis corresponding to λ_i .] In view of Exercises 24–29, explain why you expect $p(J)$ to be O , where J is a Jordan canonical form for A . Deduce that $p(A) = O$.

31. Let $T: \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a linear transformation. A subspace W of $\mathbb{C}^n$ is **invariant under T** if $T(\mathbf{w}) \in W$ for all $\mathbf{w} \in W$. Let A be the standard matrix representation of T .

- a. Describe the one-dimensional invariant subspaces of T .
- b. Show that every eigenspace E_λ of T is invariant under T .
- c. Show that the vectors in any string in a Jordan basis for A generate an invariant subspace of T .
- d. Is it true that, if S is a subspace of a subspace W that is invariant under T , then S is also invariant under T ? If not, give a counterexample.
- e. Is it true that every subspace of $\mathbb{R}^n$ invariant under T is contained in the nullspace of $(A - \lambda I)^n$, where λ is some eigenvalue of T ? If not, give a counterexample.

2. In Section 5.3, we considered systems $\mathbf{x}' = A\mathbf{x}$ of differential equations, and we saw that, if $A = CJC^{-1}$, then the system takes the form $\mathbf{y}' = J\mathbf{y}$, where $\mathbf{x} = C\mathbf{y}$. (We used D in place of J in Section 5.3, because we were concerned only with diagonalization.) Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the (not necessarily distinct) eigenvalues of an $n \times n$ matrix A , and let J be a Jordan canonical form for A .
- a. Show that the system $\mathbf{y}' = J\mathbf{y}$ is of the form

$$\begin{aligned} y_1' &= \lambda_1 y_1 + c_1 y_2, \\ y_2' &= \lambda_2 y_2 + c_2 y_3, \\ &\vdots \\ y_{n-1}' &= \lambda_{n-1} y_{n-1} + c_{n-1} y_n, \\ y_n' &= \lambda_n y_n, \end{aligned}$$

where each c_i is either 0 or 1.

- b. How can the system in part a be solved?
[HINT: Start with the last equation.]

- c. Given that, for

$$A = \begin{bmatrix} 2 & 2 & 3 \\ 0 & -1 & 0 \\ 2 & 2 & 1 \end{bmatrix},$$

$$C = \begin{bmatrix} \frac{1}{2} & 1 & \frac{3}{2} \\ 0 & -\frac{5}{4} & 0 \\ -\frac{1}{2} & 0 & 1 \end{bmatrix},$$

$$J = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{bmatrix},$$

we have $C^{-1}AC = J$, find the solution of the differential system $\mathbf{x}' = A\mathbf{x}$.

33. Let A be an $n \times n$ matrix with eigenvalue λ . Prove that the algebraic multiplicity of λ is at least as large as its geometric multiplicity.

SOLVING LARGE LINEAR SYSTEMS

The Gauss and Gauss–Jordan methods presented in Chapter 1 are fine for solving very small linear systems with pencil and paper. Some applied problems—in particular, those requiring numerical solutions of differential equations—can lead to very large linear systems, involving thousands of equations in thousands of unknowns. Of course, such large linear systems must be solved by the use of computers. That is the primary concern of this chapter. Although a computer can work tremendously faster than we can with pencil and paper, each individual arithmetic operation does take some time, and additional time is used whenever the value of a variable is stored or retrieved. In addition, indexing in subscripted arrays requires time. When solving a large linear system with a computer, we must use as efficient a computational algorithm as we can, so that the number of operations required is as small as practically possible.

We begin this chapter with a discussion of the time required for a computer to execute operations and a comparison of the efficiency of the Gauss method including back substitution with that of the Gauss–Jordan method.

Section 10.2 presents the LU (lower- and upper-triangular) factorization of the coefficient matrix of a square linear system. This factorization will appear as we develop an efficient algorithm for solving, by repeated computer runs, many systems all having the same coefficient matrix.

Section 10.3 deals with problems of roundoff and discusses ill-conditioned matrices. We will see that there are actually very small linear systems, consisting of only two equations in two unknowns, for which good computer programs may give incorrect solutions.

10.1

CONSIDERATIONS OF TIME

Timing Data for One PC

The computation involved in solving a linear system is just a lot of arithmetic. Arithmetic takes time to execute even with a computer, although the computer is obviously much faster than an individual working with pencil and paper. In reducing a matrix by using elementary row operations, we spend most of our time adding a multiple of a row vector to another row vector. A typical step in multiplying row k of a matrix A by r and adding it to row i consists of

$$\text{replacing } a_{ij} \text{ by } a_{ij} + ra_{kj}. \quad (1)$$

In a computer language such as BASIC, operation (1) might become

$$A(I,J) = A(I,J) + R*A(K,J). \quad (2)$$

Computer instruction (2) involves operations of addition and multiplication, as well as indexing, retrieving values, and storing values. We use terminology of C. B. Moler and call such an operation a **flop**. These flops require time to execute on any computer, although the times vary widely depending on the computer hardware and the language of the computer program. When we wrote the first edition of this text in 1986, we experimented with our personal computer, which was very slow by today's standards, and created the data in Table 10.1. Although computers are much faster today, the indication in the table that computation does require time remains as valid now as it was then. We left the original 1986 data in the table so that you can see, if you work Exercise 22, how much faster computers are today. To form the table, we generated two random numbers, R and S , and then found the time required to add them 10,000 times, using the BASIC loop

$$\text{FOR } I = 1 \text{ TO } N: C = R + S: \text{NEXT} \quad (3)$$

where we had earlier set $N = 10,000$. We obtained the data shown in the top row of Table 10.1. We also had the computer execute loop (3) with $+$ replaced by $-$ (subtraction), then by $*$ (multiplication), and finally by $/$ (division). We similarly timed the execution of 10,000 flops, using

$$\text{FOR } I = 1 \text{ TO } N: A(K,J) = A(K,J) + R * A(M,J): \text{NEXT}, \quad (4)$$

where we had set $N = 10,000$ and had also assigned values to all other variables except I . All our data are shown in Table 10.1 on the next page.

Table 10.1 gives us quite a bit of insight into the PC that generated the data. Here are some things we noticed immediately.

Point 1 Multiplication took a bit, but surprisingly little, more time than addition.

TABLE 10.1
Time (in Seconds) for Executing 10,000 Operations

Routine	Interpretive BASIC		Compiled BASIC	
	Single-Precision	Double-Precision	Single-Precision	Double-Precision
Addition [using (3)]	37	44	8	9
Subtraction [using (3) with -]	37	48	8	9
Multiplication [using (3) with *]	39	53	9	11
Division [using (3) with /]	47	223	9	15
Flops [using (4)]	143	162	15	18

Point 2 Division took the most time of the four arithmetic operations. Indeed, our PC found double-precision division in interpretive BASIC very time-consuming. We should try to minimize divisions as much as possible. For example, when computing

$$\left(\frac{4}{3}\right)(3, 2, 5, 7, 8) + (-4, 11, 2, 1, 5)$$

to obtain a row vector with first entry zero, we should *not* compute the remaining entries as

$$\left(\frac{4}{3}\right)2 + 11, \quad \left(\frac{4}{3}\right)5 + 2, \quad \left(\frac{4}{3}\right)7 + 1, \quad \left(\frac{4}{3}\right)8 + 5,$$

which requires four divisions. Rather, we should do a single division, finding $r = \frac{4}{3}$, and then computing

$$2r + 11, \quad 5r + 2, \quad 7r + 1, \quad 8r + 5.$$

Point 3 Our program ran much faster in compiled BASIC than in interpretive BASIC. In the compiled version on our PC, the time for double-precision division was not so far out of line with the time for other operations.

Point 4 The indexing in the flops required significant time in interpretive BASIC on our PC.

The routine TIMING in LINTEK was used to generate the data in Table 10.1 on our PC. Exercise 22 asks students to obtain analogous data with their PCs, using this program.